SYED RUMAN AHMAD

Computer Science Graduate · AI & Machine Learning
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PROFESSIONAL SUMMARY

Computer Science graduate with end-to-end experience in Machine Learning, Natural Language Processing, and applied AI development. Demonstrated ability to build and deploy production-ready solutions — from fine-tuning transformer models (BERT, DistilGPT2) and integrating large language model APIs (GPT-3.5, Mistral-7B) to delivering interactive applications via Streamlit. Brings a strong foundation in both research-oriented thinking and practical engineering, with a defined interest in the safety and reliability of LLM-based systems. Pursuing a Master's degree to advance research contributions at the intersection of NLP and applied AI.

EDUCATION

Bachelor of Science in Computer Science · University of Education, Lahore — Multan Campus *Sep 2021 – Sep 2025*

- **CGPA: 3.23 / 4.0** · Admitted on Open Merit
- Core coursework: Artificial Intelligence · Machine Learning · Cloud Computing · Computer Networks · Parallel & Distributed Computing · Data Structures & Algorithms · Database Systems · Information Security

PROFESSIONAL EXPERIENCE

AI & Machine Learning Intern · Developers Hub Corporation *Dec 2025 – Jan 2026*

- Delivered end-to-end ML solutions across multiple concurrent projects, covering data collection, preprocessing, model development, performance evaluation, and deployment
- **Transformer fine-tuning:** fine-tuned BERT (bert-base-uncased) for multi-class news topic classification and DistilGPT2 on the EmpatheticDialogues dataset using Hugging Face Transformers and PyTorch
- **LLM integration:** developed a domain-specific health-query chatbot by integrating GPT-3.5 and Mistral-7B APIs; engineered custom prompt templates and safety filters to constrain outputs to accurate, reliable responses
- **Predictive modelling:** built and benchmarked Linear Regression and Random Forest models for short-term stock price prediction on time-series data; evaluated using F1-score, Accuracy, MAE, and RMSE
- **Deployment:** packaged and deployed all AI projects as interactive Streamlit web applications, enabling live model testing and stakeholder demonstration

RESEARCH INTERESTS & PROPOSAL

Core research interest: trustworthy and explainable AI, with a focus on NLP and healthcare applications. Internship experience with health-query LLM systems highlighted a persistent gap — general-purpose large language models produce unreliable or unsafe outputs in high-stakes domains when used by non-expert users. The proposed research direction investigates domain-specific fine-tuning strategies and Retrieval-Augmented Generation (RAG) pipelines as mechanisms to improve factual accuracy, output safety, and contextual grounding of LLM-based medical assistants — with particular emphasis on deployment in resource-constrained settings. This work directly contributes to the growing research agenda on responsible AI and human-centred machine learning.

TECHNICAL PROJECTS

News Topic Classifier | *BERT · Hugging Face Transformers · PyTorch · Streamlit*

- Fine-tuned bert-base-uncased on a multi-class news dataset; achieved reliable classification performance and deployed the model as an interactive Streamlit application for real-time inference

LLM Health Chatbot | *GPT-3.5 API · Mistral-7B API · Prompt Engineering · Safety Filtering*

- Designed a domain-constrained conversational AI system using GPT-3.5 and Mistral-7B; implemented structured prompt templates and output safety filters to ensure accurate, context-aware health responses

Empathetic Dialogue Generator | *DistilGPT2 · Hugging Face Transformers · EmpatheticDialogues Dataset*

- Fine-tuned DistilGPT2 on the EmpatheticDialogues benchmark dataset; applied prompt conditioning techniques to improve response tone alignment and contextual coherence across conversation turns

Short-Term Stock Price Predictor | *Python · Scikit-learn · yfinance · Matplotlib*

- Engineered time-series features from financial data sourced via yfinance; trained and compared Linear Regression and Random Forest models for next-day closing price prediction, with visualised actual-vs-predicted output for model interpretability

Secure Management Portal | *Node.js · Express.js · MySQL · RBAC — Final Year Project*

- Architected and delivered a full-stack web application serving 30+ users; implemented Role-Based Access Control (RBAC) for granular permission enforcement and secure file upload functionality

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, HTML/CSS

Machine Learning: Scikit-learn, Pandas, NumPy, Regression, Classification, Random Forest, Time-Series Analysis

NLP & Large Language Models: BERT, DistilGPT2, GPT-3.5 API, Mistral-7B API, Hugging Face Transformers, PyTorch, Prompt Engineering, Model Fine-tuning, Retrieval-Augmented Generation (RAG)

Web & Deployment: Streamlit, Node.js, Express.js, MySQL

Tools: Hugging Face Hub, yfinance, Git, GitHub

CERTIFICATIONS

- ChatGPT Prompt Engineering for Developers — DeepLearning.AI (2024)
- Introduction to Machine Learning — Kaggle (2024)
- AI for Social Impact — ADBI, Asian Development Bank Institute (2024)